

MARVIN WONG

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EDUCATION

University of California, Los Angeles (UCLA)

Sep 2025 - Dec 2026

Master of Engineering, Data Science

- **GPA:** 4.0
- **Coursework:** Neural Networks and Deep Learning, Large-Scale Machine Learning, Product Management, Financial Management, Advanced Natural Language Processing

University of California, Los Angeles (UCLA)

Sep 2021 - Jun 2025

Bachelors of Science, Data Theory, Atmospheric and Oceanic Sciences

- **GPA:** 3.83
- **Coursework:** Statistical Modeling, Advanced Linear Algebra, Machine Learning, Data Structures and Algorithms, Data Analytics, Design and Analysis of Experiment, Probability, Statistical Models in Finance

TECHNICAL SKILLS

- **Languages and Core:** Python, SQL, R, Bash, Excel
- **Frameworks, Packages, Cloud:** PyTorch, TensorFlow, Scikit-Learn, Pandas, NumPy, Apache Spark, Databricks, AWS, R, Docker, Tableau, GCP, FastAPI, Matplotlib, MCP, LangChain

EXPERIENCE

UCLA Computer Science Department

Mar 2026 - Present

Graduate Research Assistant

- Designed a retrieval and matching system utilizing geometric analysis, increasing model success rates from 70% to 90%
- Engineered an automated machine learning preprocessing workflow using vision-based embedding models and **PyTorch** to segment and organize large-scale datasets for downstream neural network training

Hong Kong University of Science and Technology

Jun 2024 - Sep 2024

Research Assistant - Atmospheric Data Science

- Built a Python pipeline that scraped and consolidated **3+ GB** of NOAA climate data, using **Pandas** for cleaning and **Scikit-Learn** for preprocessing to streamline research team analysis
- Developed and optimized a topological time-series forecasting model (TDA) for climate signals, achieving a **50%** performance improvement over baseline methods

PricewaterhouseCoopers

Jul 2023 - Sep 2023

Data Analyst Intern

- Designed executive-level **Tableau** visualizations and reports to translate KPIs into actionable insights for key client stakeholders
- Conducted **A/B testing** experiments on UI features, utilizing statistical hypothesis testing to optimize user engagement (+15%) and daily transaction volume (+15%) on SAP legacy systems
- Facilitated 4 company-wide training workshops to onboard teams, securing an 80% platform adoption rate

Hysan Development Company

Intern

- Modeled foot traffic and financial return scenarios in **Excel** (VLOOKUP, Pivot Tables) for commercial proposals, reducing weekly reconciliation work by 2 hours and supporting executive investment decisions

FEATURED PROJECTS

Securiport Visual Intelligence Capstone

- Engineered an end-to-end multimodal machine learning pipeline using **PyTorch**, deploying state-of-the-art vision models (- **YOLOv8**, **DINOv2**) for object detection and feature extraction and achieving 98.40% track-level identification accuracy
- Built a vector-based search engine in **NumPy** and **Scikit-learn** using cosine similarity and multi-object tracking to maintain target classification across continuous video streams

Reddit r/AITA Language Model & API Deployment

- Fine-tuned a **RoBERTa** transformer model (**Hugging Face**, **PyTorch**) on 110k+ unstructured text samples, improving Macro F1 score from 0.46 to 0.71+ (54% improvement over TF-IDF baselines)
- **Performed deep-dive statistical analysis on user interaction data**, utilizing point-biserial correlations and Chi-squared tests on engagement metadata (comment-to-score ratios) to isolate key behavioral signals
- Containerized the model using **Docker** and deployed a high-throughput **FastAPI** REST service on **GCP Cloud Run**

BirdCLEF+ Audio Classifier

- Achieved **0.86 test accuracy**, placing in the top 10% of 2,000 submissions, on passive acoustic monitoring data from rainforest environments
- Applied pseudo-labeling on unlabeled audio to mitigate domain shift and improve generalization

ACCOMPLISHMENTS

- **2023 American Statistical Association DataFest Finalist:** Top 12 finalist out of 100 teams; conducted NLP sentiment analysis and statistical regression on American Bar Association appointment transcripts in Python to address customer concerns and backlogs